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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm

CSA1511 - CLOUD COMPUTING AND BIG DATA ANALYTICS

Technical Presentation

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INTRODUCTION

What is Edge Computing?

Edge Computing is a distributed computing paradigm that processes data near the source where it is generated, such as IoT devices, sensors, or edge servers, instead of relying solely on centralized cloud data centers.

What is Cloud Computing?

Cloud Computing is a technology that provides computing resources such as storage, processing power, networking, and software services over the internet through centralized data centers on an on-demand basis.

Why Compare Edge and Cloud Computing?

Modern applications require both high computational power and real-time responsiveness.

Cloud Computing offers centralized processing and virtually unlimited scalability.

Edge Computing reduces latency by processing data closer to end devices.

Selecting the appropriate computing paradigm depends on factors such as application requirements, response time, and network conditions.



CORE TECHNICAL EXPLANATION

Working of Cloud Computing

- Data is generated by user devices or IoT sensors.
- Information is transmitted through the Internet.
- Cloud servers store and process the data.
- Results are sent back to users or applications.
- Suitable for large-scale storage and complex analytics.

Working of Edge Computing

- Data is processed locally at the edge device or nearby edge server.
- Only necessary information is forwarded to the cloud.
- Enables faster response and real-time decision making.
- Reduces bandwidth usage and network congestion.

Working Process

Device/Sensor → Edge Node → Cloud Server (Optional) → User Application



Edge and Cloud Architecture

Components of Edge Computing

1. Edge Devices

Sensors

Cameras

IoT Devices

Smart Phones

2. Edge Gateway

Local Data Processing

Data Filtering

Device Management



Edge and Cloud Architecture

Components of Edge Computing

3. Communication Network

Wi-Fi

5G

Ethernet

Bluetooth

4. Cloud Platform

Long-term Storage

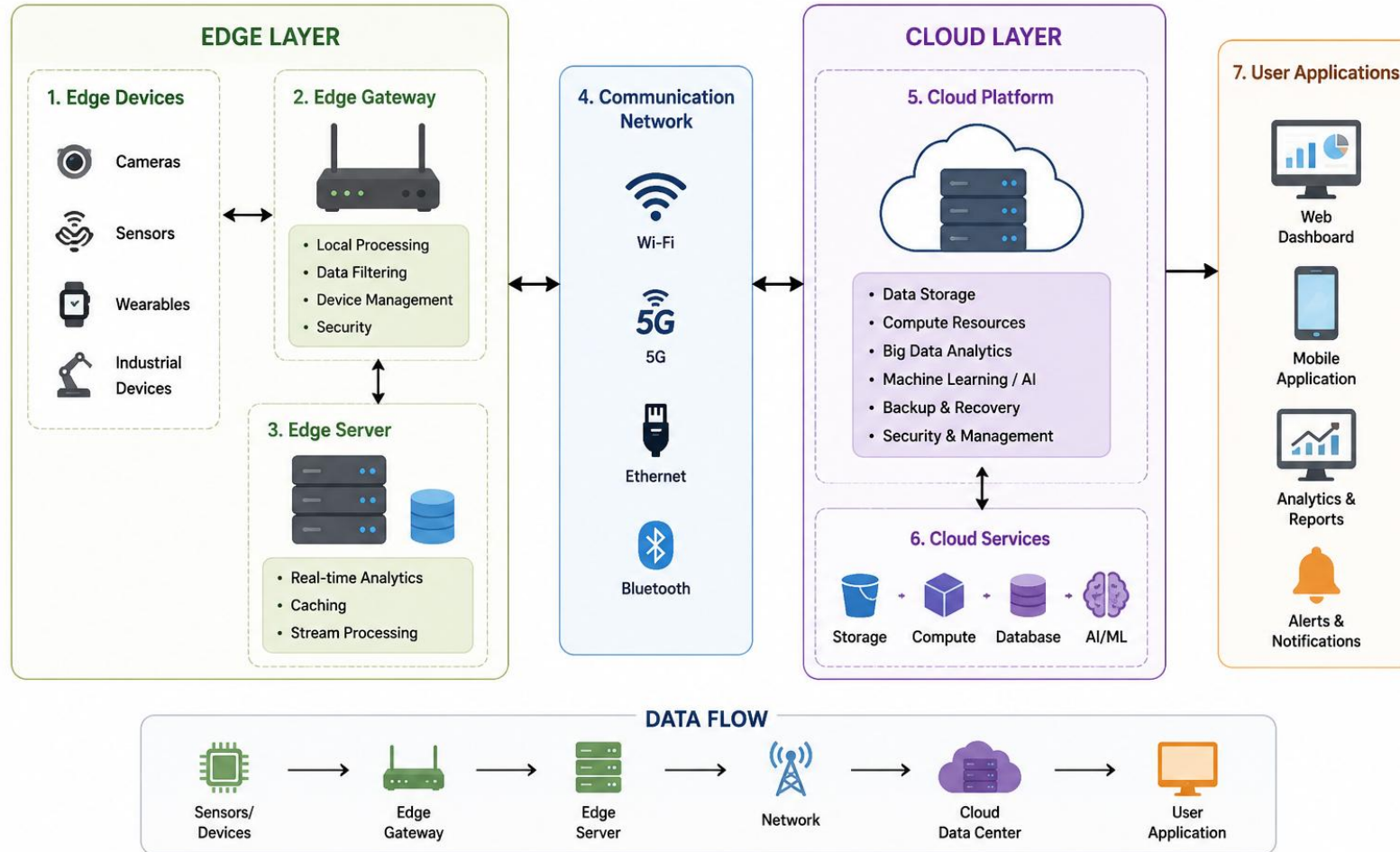
Big Data Analytics

AI & Machine Learning



ARCHITECTURE DIAGRAM

EDGE-CLOUD ARCHITECTURE DIAGRAM





COMPARISON

Edge Computing vs Cloud Computing

Edge Computing	Cloud Computing
Processing near devices	Processing in centralized servers
Very low latency	Higher latency
Faster response	Depends on network speed
Lower bandwidth usage	Higher bandwidth usage
Limited storage	Virtually unlimited storage
Better for real-time applications	Better for large-scale analytics



REAL-WORLD APPLICATIONS

Edge Computing

- Autonomous Vehicles
- Smart Manufacturing
- Video Surveillance
- Smart Cities
- Healthcare Monitoring

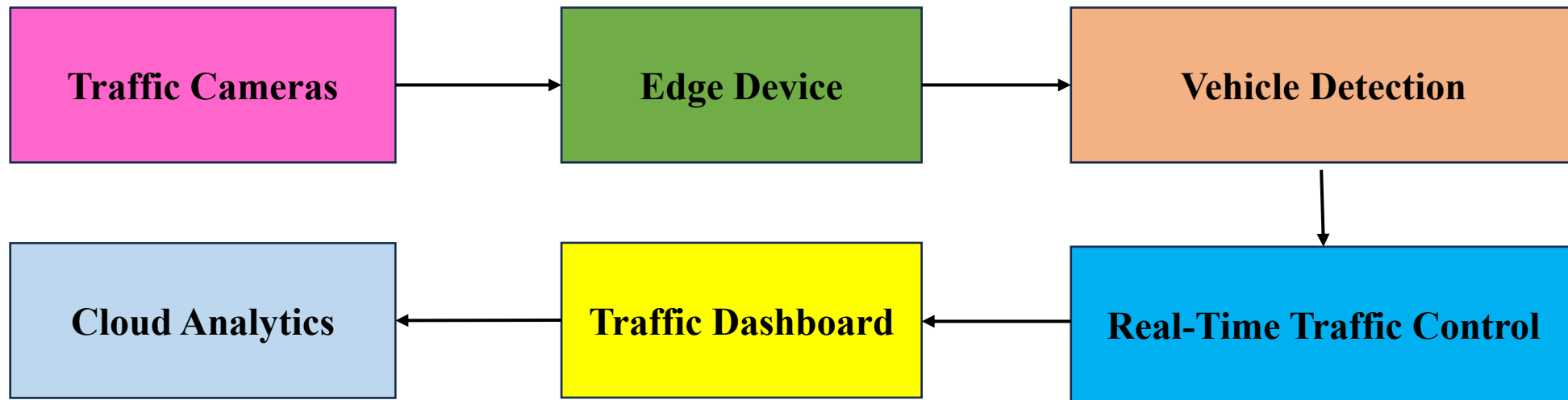
Cloud Computing

- Online Storage
- Video Streaming
- Business Applications
- Big Data Analytics
- Artificial Intelligence



EXAMPLE

Example Use Case Smart Traffic Management System



Advantages

- Faster traffic monitoring
- Reduced network congestion
- Low latency decision making
- Improved road safety



CONCLUSION

- Edge Computing and Cloud Computing are complementary technologies rather than competitors.
- Edge Computing enables low-latency processing near data sources.
- Cloud Computing provides scalable storage, computing power, and advanced analytics.
- A hybrid Edge-Cloud model delivers the best performance for modern applications.

THANK YOU